Ex : 02
5/2/26

DETERMINATION OF CURRENT USING MESH ANALYSIS.

AIM :

To determine the current in circuit using mesh analysis both theoretically and practically for a given DC circuit.

APPARATUS REQUIRED :

SL. NO	APPARATUS	SPECIFICATION	QUANTITY
1.	Regulated Power supply (RPS)	(0-30V)	1
2.	Ammeter	(0-10 mA) MC	3
3.	Resistors	330Ω, 470Ω, 560Ω	Each 1
4.	Bread board	—	1

PROCEDURE :

* Given connections as per the circuit diagram.

* Switch ON the supply, vary the RPS (Regulated Power Supply) and set a particular input voltage.

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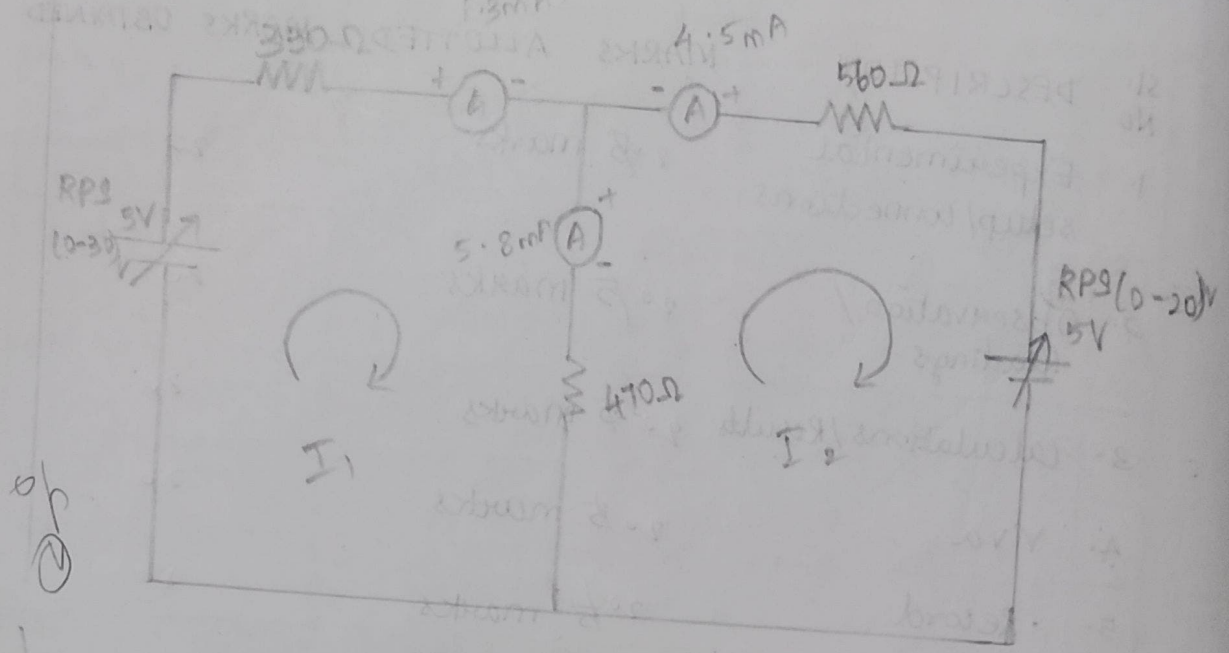
CIRCUIT DIAGRAM:



MESH - CIRCUIT

TABULAR COLUMN

PARAMETERS	THEORETICAL	PRACTICAL
330Ω	1.2mA	1.3mA
560Ω	4.2mA	4.3mA
470Ω	5.4mA	5.4mA

CALCULATIONS :

$$-3 + 330 I_1 + 470 (I_1 - I_2) = 0$$

$$330 I_1 + 470 I_1 - 470 I_2 = 3$$

$$800 I_1 - 470 I_2 = 3 \longrightarrow (1)$$

$$560 I_2 + 5 + 470 (I_2 - I_1) = 0$$

$$560 I_2 + 470 I_2 - 470 I_1 = -5$$

$$-470 I_1 + 1030 I_2 = 5 \rightarrow (2)$$

By cramer's rule for (1) and (2)

$$\begin{bmatrix} 800 & -470 \\ -470 & 1030 \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \end{bmatrix} = \begin{bmatrix} 3 \\ 5 \end{bmatrix}$$

$$\Delta = 800 (1030) - (470)(470)$$

$$= 824000 - 220400$$

$$\boxed{\Delta = 603100}$$

$$\Delta_1 = \begin{bmatrix} 3 & -470 \\ -5 & 1030 \end{bmatrix} = 3090 - 2350$$

$$\boxed{\Delta_1 = 740}$$

$$\Delta_2 = \begin{bmatrix} 800 & 3 \\ -470 & -5 \end{bmatrix} = -4000 + 1410$$

$$\boxed{\Delta_2 = -2590}$$

$$I_1 = \frac{\Delta_1}{\Delta} = \frac{740}{603100} = 0.0012A$$

$$\boxed{I_1 = 1.2\,mA}$$

$$I_2 = \frac{\Delta_2}{\Delta} = \frac{-2590}{603100} = -0.0042A$$

$I_2 = -4.2\,mA$ (-ve sign indicates opposite direction)

$$I_3 = I_1 + I_2$$

$$= 1.4 + 4.2$$

$$\boxed{I_3 = 5.4\,mA}$$

* Note down the readings of ammeters and voltmeters and tabulate them.

* Vary the RPS for different input voltage and note down the readings of all the meters.

* Reduce the RPS to its minimum value and switch OFF the supply.

* Using the tabulated values, verify Kirchoff's laws practically, and verify it theoretically.

MARKS SPLIT-UP :

SL. NO	DESCRIPTION	MARKS ALLOTTED	MARKS OBTAINED
1.	Experimental setup / Connections	5 marks	20
2.	Observation /Readings	5 marks	20
3.	Calculations /Results	5 marks	20
4.	Viva	5 marks	20
5.	Record.	5 marks	20
		Total	100

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RESULT :

Thus, the Mesh Analysis is verified practically and theorectically. The resultant currents for 1V supply are,

a) The current across 330Ω resistor is ~~0.0012A~~ 1.2 mA

b) The current across 560Ω resistor is 4.2 mA.

c) The current across 470Ω resistor is 5.4 mA.